

3T IN ONCOLOGY COMPETITION PROJECT PRE-EVALUATION FORM

**PROJECT NAME: AI-Driven Subtype-Specific
Drug Repurposing Platform for GBM**

TEAM NAME: KBU-MedLab

APPLICATION ID:5009318

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1. BIOTECHNOLOGY FIELD

This project falls under the categories of Genetic/Epigenetic Approaches, Medical Oncology, Medical Genetics, and Artificial Intelligence Supported Next Generation Drug Development Solutions. The objective of this project is Glioblastoma (GBM), the most aggressive brain tumour. To identify tumor subtypes, biomarkers, therapeutic targets, and drug repurposing opportunities, our suggested approach combines artificial intelligence, bioinformatics, and comprehensive genomic analysis.

2. PROJECT SUMMARY

GBM is not a single disease, it's a collection of molecularly distinct subtypes, each behaving differently and responding to treatment in its own way. That's the core idea behind KBU-MedLab's computational precision oncology platform. We will be working with four sequential engines integrated into the platform: a Subtype Engine for tumour clustering, a Biomarker Engine for finding gene signatures unique to subtypes, a Target Discovery Engine that works to uncover meaningful therapeutic targets within each subtype, and a Drug Repurposing Engine. In order to find significant patterns, the system evaluates patient tumour data obtained via RNA-seq gene expression profiling. It identifies therapeutic weaknesses, describes the behaviour and level of aggressiveness of tumours, finds important genes that influence development, and suggests possible drugs that target important pathways. Validated GBM subtype classifications, biomarker signatures, prioritised druggable targets, and ranking repurposable treatments for each subtype are among the expected outcomes. By doing all of these parts, the platform functions as an end-to-end decision-support system that changes raw biological data into insightful knowledge, enabling glioblastoma patients to receive more precise and targeted treatment regimens.

3-PROBLEM DEFINITION

The most severe primary brain tumour, glioblastoma multiforme (GBM) was selected due to its significant resistance to treatment, invasive nature, and rapid development (Tan et al., 2020). The fact that GBM is treated as a single illness despite the fact that it in fact comprises several molecular subtypes with unique genetic profiles and drug responses is a significant challenge (Verhaak et al., 2010). Treatment is challenging due to this diversity, leading to resistance and poor outcomes (Patel et al., 2014). Current approaches do not entirely address such problems, and developing new medications is difficult and time consuming. Therefore, in order to determine GBM subtypes, identify key targets, and match them with effective treatments, data-driven approaches are required.

3. Solution

The KBU-MedLab platform addresses GBM through a four-engine pipeline, where each step builds on the last. We start with the Subtype Engine, which runs unsupervised clustering on

RNA-seq gene expression data from TCGA-GBM, using STAR-Counts-processed data limited to primary tumours. From there, the Biomarker Engine runs a one-vs-rest differential expression analysis for each subtype — essentially comparing one subtype against all the others to pin down the gene signatures that make each one distinct. This is where we start to understand the molecular identity of each subtype and where its vulnerabilities might lie. The Target Discovery Engine then takes those biomarkers further by bringing in functional genomics data from DepMap, specifically CRISPR-Cas9 and RNAi loss-of-function screens, to figure out which genes tumour cells actually depend on to survive. Finally, the Drug Repurposing Engine takes those critical targets and cross-references them against DGIdb and DrugBank to find existing drugs that could realistically be redirected toward GBM. What makes this pipeline different is that all of these steps are connected — transcriptomics, functional genomics, and pharmacological data all feeding into one another, so we can move from subtype identification to candidate treatments without starting from scratch with new drug development.

4. INNOVATIVE ASPECT AND UNIQUE VALUE

Earlier studies have investigated aspects of this tumour. Soyer et al. (2025) looked at combining biomarker discovery with drug repurposing, and Zhao et al. (2023) used DepMap CRISPR data to find essential genes in GBM. There's also been growing interest in applying explainable AI to glioma research. But these efforts have largely stayed separate. What KBU-MedLab is doing differently is pulling all of it together subtype classification, target validation, drug repurposing, and explainable AI into one connected pipeline. We're not just making predictions; we're making sure those predictions can actually be understood and interpreted, which matters a lot when the goal is eventually clinical use.

5. TECHNOLOGY READINESS LEVEL (TRL):

At this stage, the platform sits at TRL 2, meaning the concept is fully formed but not yet implemented. We've mapped out the four-engine architecture, chosen our data sources — TCGA-GBM through the GDC Data Portal and DepMap for CRISPR/RNAi dependency data — and defined the analytical approach for each engine. The theoretical groundwork has been laid through a review of published literature on GBM subtypes, gene dependency scoring, and computational drug repurposing. That said, no code has been written, no data has been run through the system, and no prototype exists yet. We will not be using any real samples in our training. We're at the ideas-on-paper stage, which puts us well within the TRL 2–3 range as required by the competition.

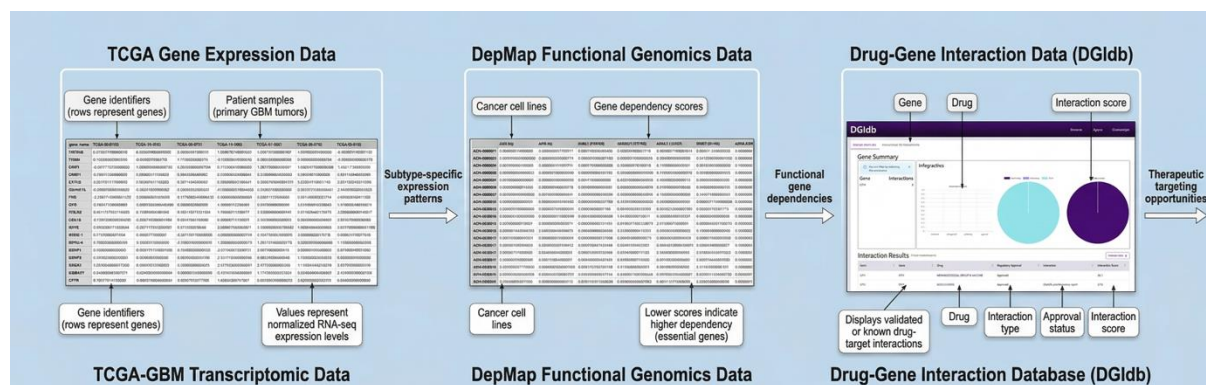


Figure 1. Data Sources and Integrated Analysis Pipeline for GBM

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